DROSE IGNATIOUS SHANE

Cryo-EM Facility Manager
Bangalore, India | +91 95978 92427 | ignatious9894@gmail.com
linkedin.com/in/drose-ignatious-shane | ORCID: orcid.org/0000-0002-5329-0748

85+ EMDB structures | **30+** active users | **2.8 Å** best resolution | **95%** uptime | **100+** platform users

PROFESSIONAL SUMMARY

Cryo-EM Facility Manager at IISc Bangalore, running a national facility with the Talos Arctica (200 kV) and Talos L120C for academic and industry users. Combine hands-on expertise in vitrification, SPA and Cryo-ET with user training, vendor coordination and maintenance, and a Django-based web platform I built for booking, billing and reporting.

PROFESSIONAL EXPERIENCE

Cryo-EM Facility Manager | Indian Institute of Science (IISc), Bangalore **2023 – Present**

- **Operations:** Lead day-to-day running of a national Cryo-EM facility serving **30+ active users across academia and industry**, supporting **85+ EMDB-deposited structures**, with data reaching **2.8 Å resolution**.
- **Instruments:** Manage the Talos Arctica (200 kV) and Talos L120C at **95% uptime**. Coordinate Thermo Fisher and Gatan service and contracts, carry out preventive maintenance twice a year, and troubleshoot ice contamination, stage drift, autoloader errors and beam misalignment to keep user sessions on track.
- **Sample prep & user support:** Screen grids and optimise vitrification (Vitrobot, VitriFlex, Leica EM GP2). Trained and supported 50+ academic users and helped researchers with 3D reconstruction, map validation and high-resolution structure determination (RELION, CryoSPARC, EMAN), working with faculty and students on experiment design.
- **Facility platform:** Built and maintain a facility management platform (Django) for 100+ registered users, covering slot booking, queue management, user records, billing, revenue reporting and an SOP/documentation repository, giving the convener real-time visibility into usage and income.
- **Administration & safety:** Manage LN2 supply, safety briefings, consumables and inventory; handle cost tracking, fee structure and reporting to the convener; manage storage, backup and transfer of user datasets.

TECHNICAL SKILLS

Cryo-EM & Structural Biology: Single-Particle Analysis (SPA) • Cryo-ET • Grid Screening & Vitrification • NS-TEM • 3D Reconstruction & Map Validation

Instruments: Talos Arctica 200 kV • Talos L120C • Vitrobot Mark IV • VitriFlex • Leica EM GP2

Software: EPU • Latitude (Gatan) • RELION • CryoSPARC • cisTEM • EMAN • UCSF Chimera / ChimeraX • Coot • Phenix • MolProbity

Facility Operations: Preventive Maintenance & Calibration • User Training & Protocols • Project Coordination • LN2 & Safety Management

Lab & Computational: Protein Purification (Ni-NTA, SEC) • Molecular Cloning • Fluorescence Microscopy • Linux • Python (basic) • Django • HTML/CSS/JavaScript • GitHub Copilot • Molecular Docking & Simulation

EDUCATION

M.Tech, Biotechnology | Kalasalingam Academy of Research and Education | GPA: 9.16/10 **2022 – 2024**
Thesis: Structural and Functional Evaluation of Mutated Alpha Hemolysin using Cryo-EM

B.Tech, Biotechnology | Kalasalingam Academy of Research and Education | GPA: 8.7/10 **2018 – 2022**
Project: Pharmacoinformatics Screening of *Aegle marmelos* Phytochemicals for IBS Targets

PUBLICATIONS

- **Shane, D. I.**, Chatterjee, A., Das, P. P., et al. A unified cryo-EM workflow for high-resolution characterization of liposomes and cargo-loaded liposomes. *Small* (2026). First author, in revision.
- Gopakumar, G., Ramesh, B., **Shane, D. I. (published as Michaelson, D. I. S.)**, Kunjiappan, S., Kabilan, S. J., Pavadai, P. In silico analysis of phytocompounds from *Aegle marmelos* against potential targets of irritable bowel syndrome. *J. Comput. Biophys. Chem.* 22(7), 863–879 (2023). doi:10.1142/S273741652350045X
- Chandran, K., **Shane, D. I.**, Zochedh, A., Sultan, A. B., Kathiresan, T. Docking simulation and ADMET prediction based investigation on the phytochemical constituents of Noni (*Morinda citrifolia*) fruit as a potential anticancer drug. *In Silico Pharmacol.* 10(1), 14 (2022). doi:10.1007/s40203-022-00130-4

TEACHING, TRAINING & VOLUNTEERING

- **Teaching Assistant**, NPTEL course "Fundamentals of Cryo-Electron Microscopy (Cryo-EM) and 3D Image Processing in Structural Biology", IISc Bangalore (2025 – 2026)
- **Student Volunteer**, EMBO Practical Course on Cryo-Electron Microscopy and 3D Image Processing (CEM3DIP), 2024
- **Student Volunteer**, Electron Microscope Society of India (EMSI) training programme, 2025

RESEARCH PROJECTS

Master's Thesis: Structural and Functional Evaluation of Mutated Alpha Hemolysin using Cryo-EM

- Purified the recombinant protein using Ni-NTA affinity chromatography and SEC.
- Performed structural evaluation using NS-TEM and Cryo-EM (SPA) and compared mutants with wild-type strains.

Cloning & Expression of Phage Proteins in *Pseudomonas aeruginosa* (M.Tech, additional work)

- Cloned phage proteins in a host system.
- Used fluorescence microscopy of phage-infected cells to identify morphological changes in the bacterial host.

Bachelor's Project: Pharmacoinformatics Screening of *Aegle marmelos* Phytochemicals for IBS Targets

- Optimised extraction of *Aegle marmelos* leaf extract using Soxhlet extraction.
- Carried out molecular docking and simulation studies of phytocompounds.

REFERENCES

Prof. Somnath Dutta – Associate Professor, Molecular Biophysics Unit, IISc Bangalore; Convener, Cryo-EM Facility (direct supervisor) | somnath@iisc.ac.in

Prof. Vidya Mangala Prasad – Assistant Professor, Molecular Biophysics Unit, IISc Bangalore (active facility user) | vmprasad@iisc.ac.in

Prof. K. Sundar – Senior Professor & Dean, School of Bio & Chemical Sciences, Kalasalingam Academy of Research and Education (academic mentor) | sundarkr@klu.ac.in